53. Boundary Scan

This MCU has boundary scan function.

The boundary scan is a serial I/O interface based on the JTAG (Joint Test Action Group, IEEE Std.1149.1 and IEEE Standard Test Access Port and Boundary-Scan Architecture).

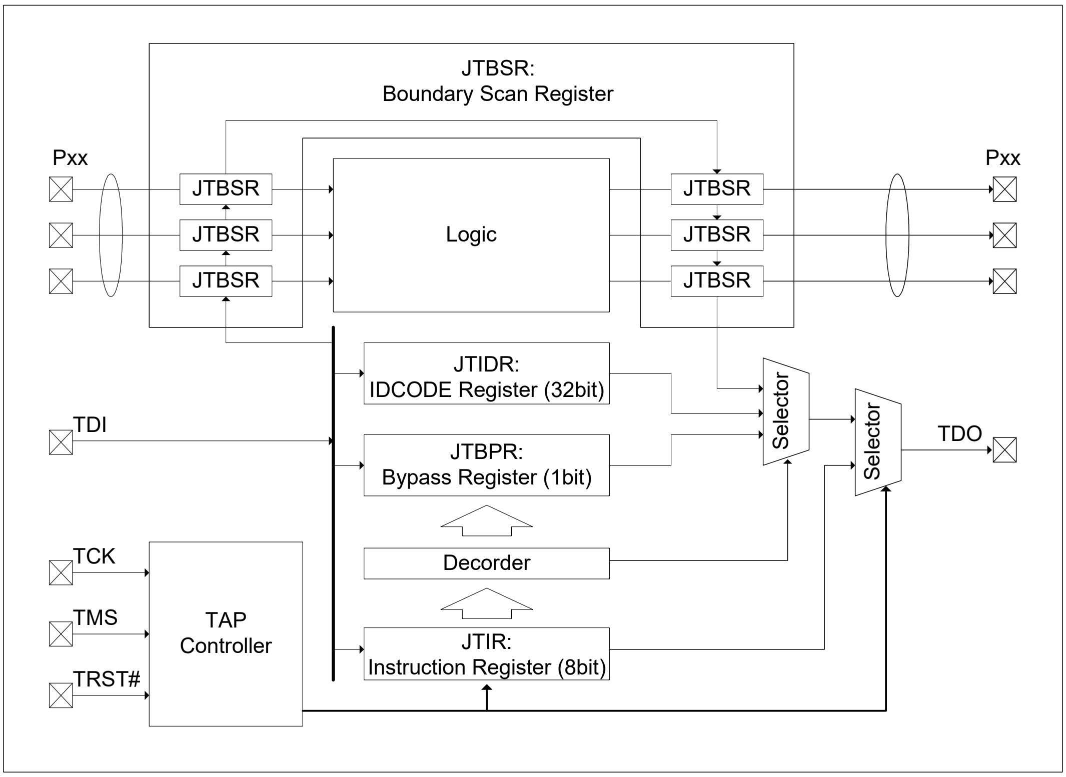
53.1 Overview

Table 53.1 lists the specifications of boundary scan.

Figure 53.1 shows a block diagram of the boundary scan function.

Table 53.1 Specifications of Boundary Scan

Item	Description
Boundary scan enabled/disabled	Boundary scan is enabled when the RES# pin and the BSCANP pin are driven high and the EMLE pin is driven low.
Dedicated boundary scan pins	The following pins are dedicated for the JTAG, when boundary scan function is enabled (TDO/TCK/TDI/TMS/TRST#). 224-pin LFBGA/176-pin LFBGA: PF0/PF1/PF2/PF3/PF4 145-pin TFLGA: P26/P27/P30/P31/P34
Six test modes	• BYPASS mode • EXTEST mode • SAMPLE/PRELOAD mode • CLAMP mode • HIGHZ mode • IDCODE mode



JTBPR: Bypass register
JTBSR: Boundary scan register
JTIR: Instruction register
JTIDR: ID code register

Figure 53.1 Block Diagram of JTAG

Table 53.2 lists the I/O pins used in the boundary scan function.

Table 53.2 Pin Configuration

Pin Name	I/O	Description
TCK	Input	Test clock input pin Clock signal for boundary scan. Input the clock the duty cycle of which is 50 percent when boundary scan function is used.
TMS	Input	Test mode select pin
TDI	Input	Test data input pin
TDO	Output	Test data output pin
TRST#	Input	Test reset input pin

53.2 Register Descriptions

Table 53.3 lists the boundary scan registers.

Table 53.3 List of Boundary Scan Registers

Register Name	Symbol	Value after Reset
Instruction register	JTIR	55h
ID code register	JTIDR	083F 9447h
Bypass register	JTBPR	Undefined
Boundary scan register	JTBSR	Undefined

Instructions can be input to the JTIR register via the TDI pin by serial transfer.
The JTBPR register, which is a 1-bit register, is connected between the TDI and TDO pins in BYPASS mode.
The JTBSR register, which is configured according to Table 53.6 to Table 53.8, is connected between the TDI and TDO pins when test data are being shifted in.
None of the registers is accessible from the CPU.
Table 53.4 lists the availability of serial transfer for the registers.

Table 53.4 Serial Transfer for the Registers

Register Name	Serial Input	Serial Output
Instruction register (JTIR)	Available	Available
ID code register (JTIDR)	Available	Available
Bypass register (JTBPR)	Available	Available
Boundary scan register (JTBSR)	Available	Available

53.2.1 Instruction Register (JTIR)

	b7	b6	b5	b4	b3	b2	b1	b0
	TS[7:0]							
Value after reset:	0	1	0	1	0	1	0	1

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	TS[7:0]	Test Bit Set	The command configuration is as listed in Table 53.5.	—

Table 53.5 Command Configuration

TS7	TS6	TS5	TS4	TS3	TS2	TS1	TS0	Instruction
0	0	0	0	0	0	0	0	EXTEST
0	1	0	0	0	0	0	0	SAMPLE/PRELOAD
0	1	0	1	0	1	0	1	IDCODE (initial value)
1	1	0	1	0	0	0	0	CLAMP
1	0	0	0	0	0	0	0	HIGHZ
1	1	1	1	1	1	1	1	BYPASS
Other than above								Reserved

JTAG instructions can be transferred to the JTIR register by serial input from the TDI pin.
The JTIR register is initialized when the TRST# pin is driven low, or when the TAP controller is in the Test-Logic-Reset state.

53.2.2 ID Code Register (JTIDR)

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	DID[31:0]															
Value after reset:	Unique value for each product*1															

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	DID[31:0]															
Value after reset:	Unique value for each product*1															

Bit	Symbol	Bit Name	Description	R/W
b31 to b0	DID[31:0]	Reserved	JTIDR is a register with the fixed value that indicates the device IDCODE.	—

Note 1. For the ID code for each product, refer to Table 53.3, List of Boundary Scan Registers.

JTIDR data is output from the TDO pin when the IDCODE instruction has been executed.

53.2.3 Bypass Register (JTBPR)

The JTBPR register is a 1-bit register and is connected between the TDI and TDO pins when the JTIR register is set to BYPASS mode.
The JTBPR register cannot be read from or written to by the CPU.

53.2.4 Boundary Scan Register (JTBSR)

The JTBSR register is a shift register to control the external input and output pins of this MCU and is distributed across the pads.
The EXTEST, SAMPLE/PRELOAD, CLAMP, and HIGHZ instructions are issued to apply the JTBSR register in boundary-scan testing.
Table 53.6 to Table 53.8 list the correspondence between the JTBSR bits and the pins of this MCU.
The value after reset is undefined.

Table 53.6 Boundary Scan Register 224-Pin LFBGA (1/12)

Pin No.	Pin Name	Input/Output	Bit Name
From TDI			
C3	P05	Output	546
		Output enable	545
		Input	544
D3	P03	Output	543
		Output enable	542
		Input	541
D6	P02	Output	540
		Output enable	539
		Input	538
D5	P01	Output	537
		Output enable	536
		Input	535
E3	P00	Output	534
		Output enable	533
		Input	532
F7	PK6	Output	531
		Output enable	530
		Input	529
F5	PK5	Output	528
		Output enable	527
		Input	526
F4	PK4	Output	525
		Output enable	524
		Input	523
G6	PF5	Output	522
		Output enable	521
		Input	520
G5	PJ5	Output	519
		Output enable	518
		Input	517
H7	PJ3	Output	516
		Output enable	515
		Input	514
G4	MD	Output	510
		Output enable	509
		Input	508
J1	EXTAL	Input	504
H2	P34	Output	503
		Output enable	502
		Input	501
H4	P33	Output	500
		Output enable	499
		Input	498

Table 53.6 Boundary Scan Register 224-Pin LFBGA (2/12)

Pin No.	Pin Name	Input/Output	Bit Name
H5	P32	Output	497
		Output enable	496
		Input	495
J4	P31	Output	488
		Output enable	487
		Input	486
J5	P30	Output	485
		Output enable	484
		Input	483
K1	PH7	Output	482
		Output enable	481
		Input	480
K2	PH6	Output	479
		Output enable	478
		Input	477
K4	PH5	Output	476
		Output enable	475
		Input	474
K3	PH4	Output	473
		Output enable	472
		Input	471
L1	PH3	Output	470
		Output enable	469
		Input	468
J6	PH2	Output	467
		Output enable	466
		Input	465
K6	PH1	Output	464
		Output enable	463
		Input	462
L2	P27	Output	455
		Output enable	454
		Input	453
M1	P26	Output	452
		Output enable	451
		Input	450
M3	P25	Output	449
		Output enable	448
		Input	447
L4	P24	Output	446
		Output enable	445
		Input	444
M2	P23	Output	443
		Output enable	442
		Input	441

Table 53.6 Boundary Scan Register 224-Pin LFBGA (3/12)

Pin No.	Pin Name	Input/Output	Bit Name
N1	P22	Output	440
		Output enable	439
		Input	438
N2	PH0	Output	437
		Output enable	436
		Input	435
P1	PK7	Output	434
		Output enable	433
		Input	432
R1	P21	Output	431
		Output enable	430
		Input	429
P3	P20	Output	428
		Output enable	427
		Input	426
P2	P17	Output	425
		Output enable	424
		Input	423
R2	P87	Output	422
		Output enable	421
		Input	420
R3	P16	Output	419
		Output enable	418
		Input	417
N3	P86	Output	416
		Output enable	415
		Input	414
J7	P15	Output	413
		Output enable	412
		Input	411
P4	P14	Output	410
		Output enable	409
		Input	408
N5	P13	Output	407
		Output enable	406
		Input	405
R4	P12	Output	404
		Output enable	403
		Input	402
L6	PJ2	Output	401
		Output enable	400
		Input	399
N6	PJ1	Output	398
		Output enable	397
		Input	396

Table 53.6 Boundary Scan Register 224-Pin LFBGA (4/12)

Pin No.	Pin Name	Input/Output	Bit Name
M5	PJ0	Output	395
		Output enable	394
		Input	393
N4	P85	Output	392
		Output enable	391
		Input	390
M6	P84	Output	389
		Output enable	388
		Input	387
P7	P57	Output	386
		Output enable	385
		Input	384
N7	P56	Output	383
		Output enable	382
		Input	381
R8	P55	Output	380
		Output enable	379
		Input	378
R7	P54	Output	377
		Output enable	376
		Input	375
P8	P11	Output	374
		Output enable	373
		Input	372
K7	P10	Output	371
		Output enable	370
		Input	369
J8	P53	Output	368
		Output enable	367
		Input	366
L8	P52	Output	365
		Output enable	364
		Input	363
M8	P51	Output	362
		Output enable	361
		Input	360
K8	P50	Output	359
		Output enable	358
		Input	357
M9	P83	Output	356
		Output enable	355
		Input	354
N9	PC7	Output	353
		Output enable	352
		Input	351

Table 53.6 Boundary Scan Register 224-Pin LFBGA (5/12)

Pin No.	Pin Name	Input/Output	Bit Name
R9	PC6	Output	350
		Output enable	349
		Input	348
R10	PC5	Output	347
		Output enable	346
		Input	345
P10	P82	Output	344
		Output enable	343
		Input	342
L9	P81	Output	341
		Output enable	340
		Input	339
N10	P80	Output	338
		Output enable	337
		Input	336
P11	PC4	Output	335
		Output enable	334
		Input	333
R11	PC3	Output	332
		Output enable	331
		Input	330
J9	PK3	Output	329
		Output enable	328
		Input	327
N11	PK2	Output	326
		Output enable	325
		Input	324
K9	PK1	Output	323
		Output enable	322
		Input	321
M10	PK0	Output	320
		Output enable	319
		Input	318
L10	PL7	Output	317
		Output enable	316
		Input	315
M12	PL6	Output	314
		Output enable	313
		Input	312
M11	PL5	Output	311
		Output enable	310
		Input	309
R12	PL4	Output	308
		Output enable	307
		Input	306

Table 53.6 Boundary Scan Register 224-Pin LFBGA (6/12)

Pin No.	Pin Name	Input/Output	Bit Name
K10	PL3	Output	305
		Output enable	304
		Input	303
P12	PL2	Output	302
		Output enable	301
		Input	300
L11	P77	Output	299
		Output enable	298
		Input	297
N12	P76	Output	296
		Output enable	295
		Input	294
P13	PC2	Output	293
		Output enable	292
		Input	291
R13	P75	Output	290
		Output enable	289
		Input	288
R14	P74	Output	287
		Output enable	286
		Input	285
M13	PM7	Output	284
		Output enable	283
		Input	282
N13	PM6	Output	281
		Output enable	280
		Input	279
R15	PM5	Output	278
		Output enable	277
		Input	276
P14	PM4	Output	275
		Output enable	274
		Input	273
P15	PM3	Output	272
		Output enable	271
		Input	270
K11	PM2	Output	269
		Output enable	268
		Input	267
J11	PN5	Output	266
		Output enable	265
		Input	264
L12	PN4	Output	263
		Output enable	262
		Input	261

Table 53.6 Boundary Scan Register 224-Pin LFBGA (7/12)

Pin No.	Pin Name	Input/Output	Bit Name
N14	PC1	Output	260
		Output enable	259
		Input	258
M14	PC0	Output	257
		Output enable	256
		Input	255
H10	P73	Output	254
		Output enable	253
		Input	252
N15	PB7	Output	251
		Output enable	250
		Input	249
J10	PL1	Output	248
		Output enable	247
		Input	246
H11	PL0	Output	245
		Output enable	244
		Input	243
M15	PB6	Output	242
		Output enable	241
		Input	240
K13	PB5	Output	239
		Output enable	238
		Input	237
L15	PB4	Output	236
		Output enable	235
		Input	234
K14	PB3	Output	233
		Output enable	232
		Input	231
L14	PB2	Output	230
		Output enable	229
		Input	228
J12	PB1	Output	227
		Output enable	226
		Input	225
K15	P72	Output	224
		Output enable	223
		Input	222
J13	P71	Output	221
		Output enable	220
		Input	219
J14	PB0	Output	218
		Output enable	217
		Input	216

Table 53.6 Boundary Scan Register 224-Pin LFBGA (8/12)

Pin No.	Pin Name	Input/Output	Bit Name
J15	PA7	Output	215
		Output enable	214
		Input	213
H14	PA6	Output	212
		Output enable	211
		Input	210
H15	PA5	Output	209
		Output enable	208
		Input	207
G15	PA4	Output	206
		Output enable	205
		Input	204
G14	PA3	Output	203
		Output enable	202
		Input	201
F15	PG7	Output	200
		Output enable	199
		Input	198
G13	PA2	Output	197
		Output enable	196
		Input	195
F14	PG6	Output	194
		Output enable	193
		Input	192
G12	PA1	Output	191
		Output enable	190
		Input	189
E14	PG5	Output	188
		Output enable	187
		Input	186
F12	PA0	Output	185
		Output enable	184
		Input	183
E15	PG4	Output	182
		Output enable	181
		Input	180
D15	P67	Output	179
		Output enable	178
		Input	177
F13	PG3	Output	176
		Output enable	175
		Input	174
C15	P66	Output	173
		Output enable	172
		Input	171

Table 53.6 Boundary Scan Register 224-Pin LFBGA (9/12)

Pin No.	Pin Name	Input/Output	Bit Name
D14	PG2	Output	170
		Output enable	169
		Input	168
C14	P65	Output	167
		Output enable	166
		Input	165
B15	PE7	Output	164
		Output enable	163
		Input	162
E13	PE6	Output	161
		Output enable	160
		Input	159
A15	P70	Output	158
		Output enable	157
		Input	156
C13	PE5	Output	155
		Output enable	154
		Input	153
B14	PE4	Output	152
		Output enable	151
		Input	150
D12	PE3	Output	149
		Output enable	148
		Input	147
F11	PM1	Output	146
		Output enable	145
		Input	144
G11	PM0	Output	143
		Output enable	142
		Input	141
H9	PN3	Output	140
		Output enable	139
		Input	138
G9	PN2	Output	137
		Output enable	136
		Input	135
H8	PQ7	Output	134
		Output enable	133
		Input	132
F9	PQ6	Output	131
		Output enable	130
		Input	129
E10	PQ5	Output	128
		Output enable	127
		Input	126

Table 53.6 Boundary Scan Register 224-Pin LFBGA (10/12)

Pin No.	Pin Name	Input/Output	Bit Name
E11	PQ4	Output	125
		Output enable	124
		Input	123
B13	PE2	Output	122
		Output enable	121
		Input	120
A14	PE1	Output	119
		Output enable	118
		Input	117
D11	PE0	Output	116
		Output enable	115
		Input	114
C11	P64	Output	113
		Output enable	112
		Input	111
B12	P63	Output	110
		Output enable	109
		Input	108
A13	P62	Output	107
		Output enable	106
		Input	105
A12	P61	Output	104
		Output enable	103
		Input	102
C10	P60	Output	101
		Output enable	100
		Input	99
A11	PD7	Output	98
		Output enable	97
		Input	96
D10	PG1	Output	95
		Output enable	94
		Input	93
B10	PD6	Output	92
		Output enable	91
		Input	90
A10	PG0	Output	89
		Output enable	88
		Input	87
D9	PD5	Output	86
		Output enable	85
		Input	84
B9	PD4	Output	83
		Output enable	82
		Input	81

Table 53.6 Boundary Scan Register 224-Pin LFBGA (11/12)

Pin No.	Pin Name	Input/Output	Bit Name
E9	PQ3	Output	80
		Output enable	79
		Input	78
G8	PQ2	Output	77
		Output enable	76
		Input	75
C9	P97	Output	74
		Output enable	73
		Input	72
A9	PD3	Output	71
		Output enable	70
		Input	69
A8	P96	Output	68
		Output enable	67
		Input	66
A7	PD2	Output	65
		Output enable	64
		Input	63
B8	P95	Output	62
		Output enable	61
		Input	60
C7	PD1	Output	59
		Output enable	58
		Input	57
B7	P94	Output	56
		Output enable	55
		Input	54
A6	PD0	Output	53
		Output enable	52
		Input	51
D7	P93	Output	50
		Output enable	49
		Input	48
A5	P92	Output	47
		Output enable	46
		Input	45
B5	P91	Output	44
		Output enable	43
		Input	42
C6	P90	Output	41
		Output enable	40
		Input	39
E8	PQ1	Output	38
		Output enable	37
		Input	36

Table 53.6 Boundary Scan Register 224-Pin LFBGA (12/12)

Pin No.	Pin Name	Input/Output	Bit Name
E7	PQ0	Output	35
		Output enable	34
		Input	33
F8	PN1	Output	32
		Output enable	31
		Input	30
E6	PN0	Output	29
		Output enable	28
		Input	27
D2	P47	Output	26
		Output enable	25
		Input	24
B4	P46	Output	23
		Output enable	22
		Input	21
D1	P45	Output	20
		Output enable	19
		Input	18
C4	P44	Output	17
		Output enable	16
		Input	15
E4	P43	Output	14
		Output enable	13
		Input	12
B3	P42	Output	11
		Output enable	10
		Input	9
A4	P41	Output	8
		Output enable	7
		Input	6
D4	P40	Output	5
		Output enable	4
		Input	3
E5	P07	Output	2
		Output enable	1
		Input	0
To TDO			

Table 53.7 Boundary Scan Register 176-Pin LFBGA (1/9)

Pin No.	Pin Name	Input/Output	Bit Name
From TDI			
B1	P05	Output	546
		Output enable	545
		Input	544
D3	P03	Output	543
		Output enable	542
		Input	541
D2	P02	Output	540
		Output enable	539
		Input	538
D1	P01	Output	537
		Output enable	536
		Input	535
D4	P00	Output	534
		Output enable	533
		Input	532
E3	PF5	Output	522
		Output enable	521
		Input	520
E1	PJ5	Output	519
		Output enable	518
		Input	517
F3	PJ3	Output	516
		Output enable	515
		Input	514
G3	MD	Output	510
		Output enable	509
		Input	508
H4	P35	Input	504
J3	P34	Output	503
		Output enable	502
		Input	501
K1	P33	Output	500
		Output enable	499
		Input	498
K2	P32	Output	497
		Output enable	496
		Input	495
L1	P31	Output	488
		Output enable	487
		Input	486
L2	P30	Output	485
		Output enable	484
		Input	483

Table 53.7 Boundary Scan Register 176-Pin LFBGA (2/9)

Pin No.	Pin Name	Input/Output	Bit Name
M1	P27	Output	455
		Output enable	454
		Input	453
M2	P26	Output	452
		Output enable	451
		Input	450
L4	P25	Output	449
		Output enable	448
		Input	447
M3	P24	Output	446
		Output enable	445
		Input	444
N2	P23	Output	443
		Output enable	442
		Input	441
N3	P22	Output	440
		Output enable	439
		Input	438
R1	P21	Output	431
		Output enable	430
		Input	429
R2	P20	Output	428
		Output enable	427
		Input	426
P2	P17	Output	425
		Output enable	424
		Input	423
P3	P87	Output	422
		Output enable	421
		Input	420
R3	P16	Output	419
		Output enable	418
		Input	417
M4	P86	Output	416
		Output enable	415
		Input	414
N4	P15	Output	413
		Output enable	412
		Input	411
P4	P14	Output	410
		Output enable	409
		Input	408
R4	P13	Output	407
		Output enable	406
		Input	405

Table 53.7 Boundary Scan Register 176-Pin LFBGA (3/9)

Pin No.	Pin Name	Input/Output	Bit Name
N5	P12	Output	404
		Output enable	403
		Input	402
M5	PJ2	Output	401
		Output enable	400
		Input	399
M6	PJ1	Output	398
		Output enable	397
		Input	396
N6	PJ0	Output	395
		Output enable	394
		Input	393
M7	P85	Output	392
		Output enable	391
		Input	390
N7	P84	Output	389
		Output enable	388
		Input	387
P7	P57	Output	386
		Output enable	385
		Input	384
R7	P56	Output	383
		Output enable	382
		Input	381
M8	P55	Output	380
		Output enable	379
		Input	378
N8	P54	Output	377
		Output enable	376
		Input	375
R8	P11	Output	374
		Output enable	373
		Input	372
P8	P10	Output	371
		Output enable	370
		Input	369
R9	P53	Output	368
		Output enable	367
		Input	366
P9	P52	Output	365
		Output enable	364
		Input	363
N9	P51	Output	362
		Output enable	361
		Input	360

Table 53.7 Boundary Scan Register 176-Pin LFBGA (4/9)

Pin No.	Pin Name	Input/Output	Bit Name
M9	P50	Output	359
		Output enable	358
		Input	357
P10	P83	Output	356
		Output enable	355
		Input	354
N10	PC7	Output	353
		Output enable	352
		Input	351
P11	PC6	Output	350
		Output enable	349
		Input	348
M10	PC5	Output	347
		Output enable	346
		Input	345
N11	P82	Output	344
		Output enable	343
		Input	342
M11	P81	Output	341
		Output enable	340
		Input	339
R12	P80	Output	338
		Output enable	337
		Input	336
P12	PC4	Output	335
		Output enable	334
		Input	333
N12	PC3	Output	332
		Output enable	331
		Input	330
M12	P77	Output	299
		Output enable	298
		Input	297
R13	P76	Output	296
		Output enable	295
		Input	294
P13	PC2	Output	293
		Output enable	292
		Input	291
P14	P75	Output	290
		Output enable	289
		Input	288
R14	P74	Output	287
		Output enable	286
		Input	285

Table 53.7 Boundary Scan Register 176-Pin LFBGA (5/9)

Pin No.	Pin Name	Input/Output	Bit Name
R15	PC1	Output	260
		Output enable	259
		Input	258
N13	PC0	Output	257
		Output enable	256
		Input	255
N14	P73	Output	254
		Output enable	253
		Input	252
M13	PB7	Output	251
		Output enable	250
		Input	249
L12	PB6	Output	242
		Output enable	241
		Input	240
M14	PB5	Output	239
		Output enable	238
		Input	237
M15	PB4	Output	236
		Output enable	235
		Input	234
L13	PB3	Output	233
		Output enable	232
		Input	231
K12	PB2	Output	230
		Output enable	229
		Input	228
L14	PB1	Output	227
		Output enable	226
		Input	225
L15	P72	Output	224
		Output enable	223
		Input	222
K13	P71	Output	221
		Output enable	220
		Input	219
K15	PB0	Output	218
		Output enable	217
		Input	216
J14	PA7	Output	215
		Output enable	214
		Input	213
J15	PA6	Output	212
		Output enable	211
		Input	210

Table 53.7 Boundary Scan Register 176-Pin LFBGA (6/9)

Pin No.	Pin Name	Input/Output	Bit Name
J12	PA5	Output	209
		Output enable	208
		Input	207
H12	PA4	Output	206
		Output enable	205
		Input	204
H13	PA3	Output	203
		Output enable	202
		Input	201
H15	PG7	Output	200
		Output enable	199
		Input	198
H14	PA2	Output	197
		Output enable	196
		Input	195
G13	PG6	Output	194
		Output enable	193
		Input	192
G14	PA1	Output	191
		Output enable	190
		Input	189
G12	PG5	Output	188
		Output enable	187
		Input	186
F14	PA0	Output	185
		Output enable	184
		Input	183
F13	PG4	Output	182
		Output enable	181
		Input	180
E15	P67	Output	179
		Output enable	178
		Input	177
E14	PG3	Output	176
		Output enable	175
		Input	174
F12	P66	Output	173
		Output enable	172
		Input	171
E13	PG2	Output	170
		Output enable	169
		Input	168
D15	P65	Output	167
		Output enable	166
		Input	165

Table 53.7 Boundary Scan Register 176-Pin LFBGA (7/9)

Pin No.	Pin Name	Input/Output	Bit Name
D14	PE7	Output	164
		Output enable	163
		Input	162
E12	PE6	Output	161
		Output enable	160
		Input	159
C15	P70	Output	158
		Output enable	157
		Input	156
D12	PE5	Output	155
		Output enable	154
		Input	153
C13	PE4	Output	152
		Output enable	151
		Input	150
B15	PE3	Output	149
		Output enable	148
		Input	147
A15	PE2	Output	122
		Output enable	121
		Input	120
A14	PE1	Output	119
		Output enable	118
		Input	117
B14	PE0	Output	116
		Output enable	115
		Input	114
B13	P64	Output	113
		Output enable	112
		Input	111
A13	P63	Output	110
		Output enable	109
		Input	108
C12	P62	Output	107
		Output enable	106
		Input	105
D11	P61	Output	104
		Output enable	103
		Input	102
A12	P60	Output	101
		Output enable	100
		Input	99
D10	PD7	Output	98
		Output enable	97
		Input	96

Table 53.7 Boundary Scan Register 176-Pin LFBGA (8/9)

Pin No.	Pin Name	Input/Output	Bit Name
B11	PG1	Output	95
		Output enable	94
		Input	93
A11	PD6	Output	92
		Output enable	91
		Input	90
C10	PG0	Output	89
		Output enable	88
		Input	87
D9	PD5	Output	86
		Output enable	85
		Input	84
B10	PD4	Output	83
		Output enable	82
		Input	81
A10	P97	Output	74
		Output enable	73
		Input	72
C9	PD3	Output	71
		Output enable	70
		Input	69
B9	P96	Output	68
		Output enable	67
		Input	66
C8	PD2	Output	65
		Output enable	64
		Input	63
D7	P95	Output	62
		Output enable	61
		Input	60
B8	PD1	Output	59
		Output enable	58
		Input	57
A8	P94	Output	56
		Output enable	55
		Input	54
C7	PD0	Output	53
		Output enable	52
		Input	51
D6	P93	Output	50
		Output enable	49
		Input	48
B7	P92	Output	47
		Output enable	46
		Input	45

Table 53.7 Boundary Scan Register 176-Pin LFBGA (9/9)

Pin No.	Pin Name	Input/Output	Bit Name
B6	P91	Output	44
		Output enable	43
		Input	42
C6	P90	Output	41
		Output enable	40
		Input	39
B5	P47	Output	26
		Output enable	25
		Input	24
A5	P46	Output	23
		Output enable	22
		Input	21
C5	P45	Output	20
		Output enable	19
		Input	18
D5	P44	Output	17
		Output enable	16
		Input	15
C4	P43	Output	14
		Output enable	13
		Input	12
A4	P42	Output	11
		Output enable	10
		Input	9
B4	P41	Output	8
		Output enable	7
		Input	6
B3	P40	Output	5
		Output enable	4
		Input	3
B2	P07	Output	2
		Output enable	1
		Input	0
To TDO			

Table 53.8 Boundary Scan Register 145-Pin TFLGA (1/8)

Pin No.	Pin Name	Input/Output	Bit Name
From TDI			
B3	P05	Output	546
		Output enable	545
		Input	544
D3	P03	Output	543
		Output enable	542
		Input	541
C2	P02	Output	540
		Output enable	539
		Input	538
D4	P01	Output	537
		Output enable	536
		Input	535
D1	P00	Output	534
		Output enable	533
		Input	532
D2	PF5	Output	522
		Output enable	521
		Input	520
E3	PJ5	Output	519
		Output enable	518
		Input	517
F3	PJ3	Output	516
		Output enable	515
		Input	514
G3	MD	Output	510
		Output enable	509
		Input	508
H4	P35	Input	504
J2	P33	Output	500
		Output enable	499
		Input	498
J3	P32	Output	497
		Output enable	496
		Input	495
L1	P25	Output	449
		Output enable	448
		Input	447
L4	P24	Output	446
		Output enable	445
		Input	444
L2	P23	Output	443
		Output enable	442
		Input	441

Table 53.8 Boundary Scan Register 145-Pin TFLGA (2/8)

Pin No.	Pin Name	Input/Output	Bit Name
M1	P22	Output	440
		Output enable	439
		Input	438
N1	P21	Output	431
		Output enable	430
		Input	429
N2	P20	Output	428
		Output enable	427
		Input	426
M2	P17	Output	425
		Output enable	424
		Input	423
N3	P87	Output	422
		Output enable	421
		Input	420
L3	P16	Output	419
		Output enable	418
		Input	417
M3	P86	Output	416
		Output enable	415
		Input	414
K4	P15	Output	413
		Output enable	412
		Input	411
N4	P14	Output	410
		Output enable	409
		Input	408
L5	P13	Output	407
		Output enable	406
		Input	405
M4	P12	Output	404
		Output enable	403
		Input	402
L6	P56	Output	383
		Output enable	382
		Input	381
N7	P55	Output	380
		Output enable	379
		Input	378
K5	P54	Output	377
		Output enable	376
		Input	375
K6	P53	Output	368
		Output enable	367
		Input	366

Table 53.8 Boundary Scan Register 145-Pin TFLGA (3/8)

Pin No.	Pin Name	Input/Output	Bit Name
L7	P52	Output	365
		Output enable	364
		Input	363
K7	P51	Output	362
		Output enable	361
		Input	360
M7	P50	Output	359
		Output enable	358
		Input	357
L8	P83	Output	356
		Output enable	355
		Input	354
N9	PC7	Output	353
		Output enable	352
		Input	351
M8	PC6	Output	350
		Output enable	349
		Input	348
L9	PC5	Output	347
		Output enable	346
		Input	345
N10	P82	Output	344
		Output enable	343
		Input	342
M9	P81	Output	341
		Output enable	340
		Input	339
K9	P80	Output	338
		Output enable	337
		Input	336
L10	PC4	Output	335
		Output enable	334
		Input	333
N11	PC3	Output	332
		Output enable	331
		Input	330
M10	P77	Output	299
		Output enable	298
		Input	297
K10	P76	Output	296
		Output enable	295
		Input	294
L11	PC2	Output	293
		Output enable	292
		Input	291

Table 53.8 Boundary Scan Register 145-Pin TFLGA (4/8)

Pin No.	Pin Name	Input/Output	Bit Name
N12	P75	Output	290
		Output enable	289
		Input	288
N13	P74	Output	287
		Output enable	286
		Input	285
M12	PC1	Output	260
		Output enable	259
		Input	258
M11	PC0	Output	257
		Output enable	256
		Input	255
L12	P73	Output	254
		Output enable	253
		Input	252
K11	PB7	Output	251
		Output enable	250
		Input	249
K12	PB6	Output	242
		Output enable	241
		Input	240
K13	PB5	Output	239
		Output enable	238
		Input	237
J11	PB4	Output	236
		Output enable	235
		Input	234
J10	PB3	Output	233
		Output enable	232
		Input	231
J12	PB2	Output	230
		Output enable	229
		Input	228
J13	PB1	Output	227
		Output enable	226
		Input	225
H10	P72	Output	224
		Output enable	223
		Input	222
H11	P71	Output	221
		Output enable	220
		Input	219
H12	PB0	Output	218
		Output enable	217
		Input	216

Table 53.8 Boundary Scan Register 145-Pin TFLGA (5/8)

Pin No.	Pin Name	Input/Output	Bit Name
H13	PA7	Output	215
		Output enable	214
		Input	213
G11	PA6	Output	212
		Output enable	211
		Input	210
G10	PA5	Output	209
		Output enable	208
		Input	207
G13	PA4	Output	206
		Output enable	205
		Input	204
F10	PA3	Output	203
		Output enable	202
		Input	201
F13	PA2	Output	197
		Output enable	196
		Input	195
F12	PA1	Output	191
		Output enable	190
		Input	189
E10	PA0	Output	185
		Output enable	184
		Input	183
E13	P67	Output	179
		Output enable	178
		Input	177
E11	P66	Output	173
		Output enable	172
		Input	171
E12	P65	Output	167
		Output enable	166
		Input	165
D10	PE7	Output	164
		Output enable	163
		Input	162
D13	PE6	Output	161
		Output enable	160
		Input	159
C12	P70	Output	158
		Output enable	157
		Input	156
D12	PE5	Output	155
		Output enable	154
		Input	153

Table 53.8 Boundary Scan Register 145-Pin TFLGA (6/8)

Pin No.	Pin Name	Input/Output	Bit Name
B13	PE4	Output	152
		Output enable	151
		Input	150
A13	PE3	Output	149
		Output enable	148
		Input	147
B12	PE2	Output	122
		Output enable	121
		Input	120
A12	PE1	Output	119
		Output enable	118
		Input	117
C11	PE0	Output	116
		Output enable	115
		Input	114
D9	P64	Output	113
		Output enable	112
		Input	111
C10	P63	Output	110
		Output enable	109
		Input	108
A11	P62	Output	107
		Output enable	106
		Input	105
B11	P61	Output	104
		Output enable	103
		Input	102
D8	P60	Output	101
		Output enable	100
		Input	99
C9	PD7	Output	98
		Output enable	97
		Input	96
A9	PD6	Output	92
		Output enable	91
		Input	90
D7	PD5	Output	86
		Output enable	85
		Input	84
B9	PD4	Output	83
		Output enable	82
		Input	81
C8	PD3	Output	71
		Output enable	70
		Input	69

Table 53.8 Boundary Scan Register 145-Pin TFLGA (7/8)

Pin No.	Pin Name	Input/Output	Bit Name
A8	PD2	Output	65
		Output enable	64
		Input	63
C7	PD1	Output	59
		Output enable	58
		Input	57
B8	PD0	Output	53
		Output enable	52
		Input	51
D6	P93	Output	50
		Output enable	49
		Input	48
A7	P92	Output	47
		Output enable	46
		Input	45
B7	P91	Output	44
		Output enable	43
		Input	42
A6	P90	Output	41
		Output enable	40
		Input	39
B6	P47	Output	26
		Output enable	25
		Input	24
C5	P46	Output	23
		Output enable	22
		Input	21
A5	P45	Output	20
		Output enable	19
		Input	18
E5	P44	Output	17
		Output enable	16
		Input	15
B5	P43	Output	14
		Output enable	13
		Input	12
A4	P42	Output	11
		Output enable	10
		Input	9
C4	P41	Output	8
		Output enable	7
		Input	6
A3	P40	Output	5
		Output enable	4
		Input	3

Table 53.8 Boundary Scan Register 145-Pin TFLGA (8/8)

Pin No.	Pin Name	Input/Output	Bit Name
A2	P07	Output	2
		Output enable	1
		Input	0
To TDO			

53.3 Operations

The boundary scan functionality is valid when the RES# pin is driven high, the EMLE pin is driven low, and the BSCANP pin is driven high.

53.3.1 TAP Controller

Figure 53.2 shows the state transition diagram of the TAP controller.

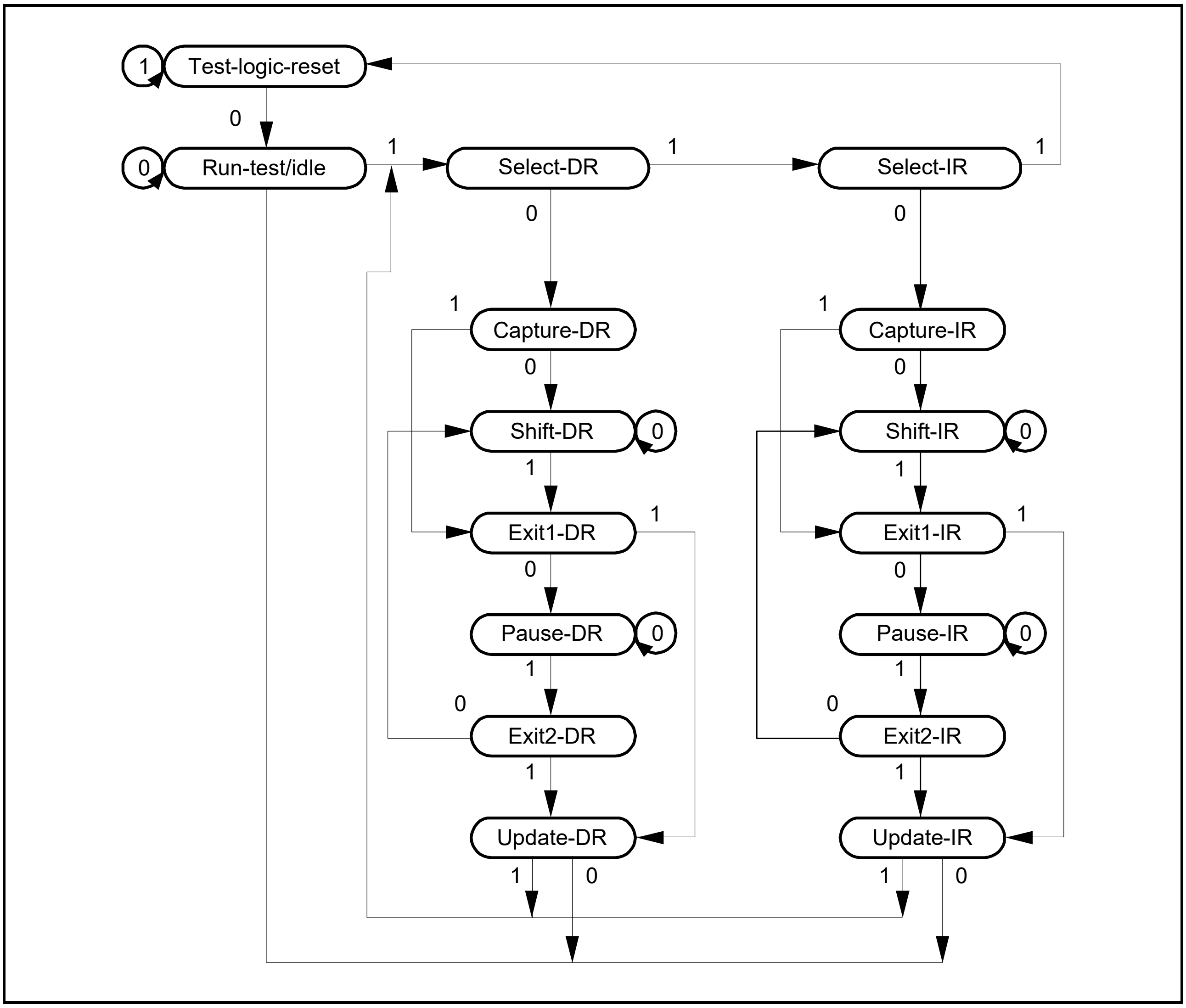


Figure 53.2 State Transition Diagram of TAP Controller

53.3.2 List of Commands

(1) BYPASS [Instruction Code: 1111 1111b]

The BYPASS instruction is an instruction that drives the bypass register (JTBPR). This instruction shortens the shift path, facilitating the transfer of serial data to other LSIs on a printed-circuit board at higher speeds. While this instruction is being executed, the test circuit has no effect on the system circuits.
The bypass register (JTBPR) is connected between the TDI and TDO pins. Bypass operation is initiated from shift-DR operation. The TDO is low in the first clock cycle in the shift-DR state; in the subsequent clock cycles, the TDI signal is output on the TDO pin.

(2) EXTEST [Instruction Code: 0000 0000b]

The EXTEST instruction is used to test external circuits when this LSI is installed on the printed circuit board. If this instruction is executed, output pins are used to output test data (specified by the SAMPLE/PRELOAD instruction) from the boundary scan register to the print circuit board, and input pins are used to input test result.

(3) SAMPLE/PRELOAD [Instruction Code: 0100 0000b]

The SAMPLE/PRELOAD instruction is used to input data from the LSI internal circuits to the boundary scan register, output data from scan path, and reload the data to the scan path. While this instruction is executed, input signals are directly input to the LSI and output signals are also directly output to the external circuits. The LSI system circuit is not affected by this instruction.
In SAMPLE operation, the boundary scan register latches the snap shot of data transferred from input pins to internal circuit or data transferred from internal circuit to output pins. The latched data is read from the scan path. The scan register latches the snap data at the rising edge of the TCK in Capture-DR state. The scan register latches snap shot without affecting the LSI normal operation.
In PRELOAD operation, initial value is written from the scan path to the parallel output latch of the boundary scan register prior to the EXTEST instruction execution. If the EXTEST is executed without executing this PRELOAD operation, undefined values are output from the beginning to the end (transfer to the output latch) of the EXTEST sequence. (In EXTEST instruction, output parallel latches are output to the output pins.)

(4) IDCODE [Instruction Code: 0101 0101b]

When the IDCODE instruction is selected, IDCODE register value is output to the TDO in Shift-DR state of the TAP controller. In this case, IDCODE register value is output from the LSB. During this instruction execution, test circuit does not affect the system circuit. JTIR is initialized by the IDCODE instruction in Test-Logic-Reset state of the TAP controller.

(5) CLAMP [Instruction Code: 1101 0000b]

When the CLAMP instruction is selected, output pins output the boundary scan register value which was specified by the SAMPLE/PRELOAD instruction in advance. While the CLAMP instruction is selected, the status of boundary scan register is maintained regardless of the TAP controller state.
BYPASS is connected between TDI and TDO, the same operation as BYPASS instruction can be achieved.

(6) HIGHZ [Instruction Code: 1000 0000b]

When the HIGHZ instruction is selected, all output pins enter high-impedance state. While the HIGHZ instruction is selected, the status of boundary scan register is maintained regardless of the state of the TAP controller.
BYPASS is connected between TDI and TDO pins, leading to the same operation as when the BYPASS instruction has been selected.

53.4 Usage Notes

(1) Pin serial transfer, data are input or output in LSB order (refer to Figure 53.3).

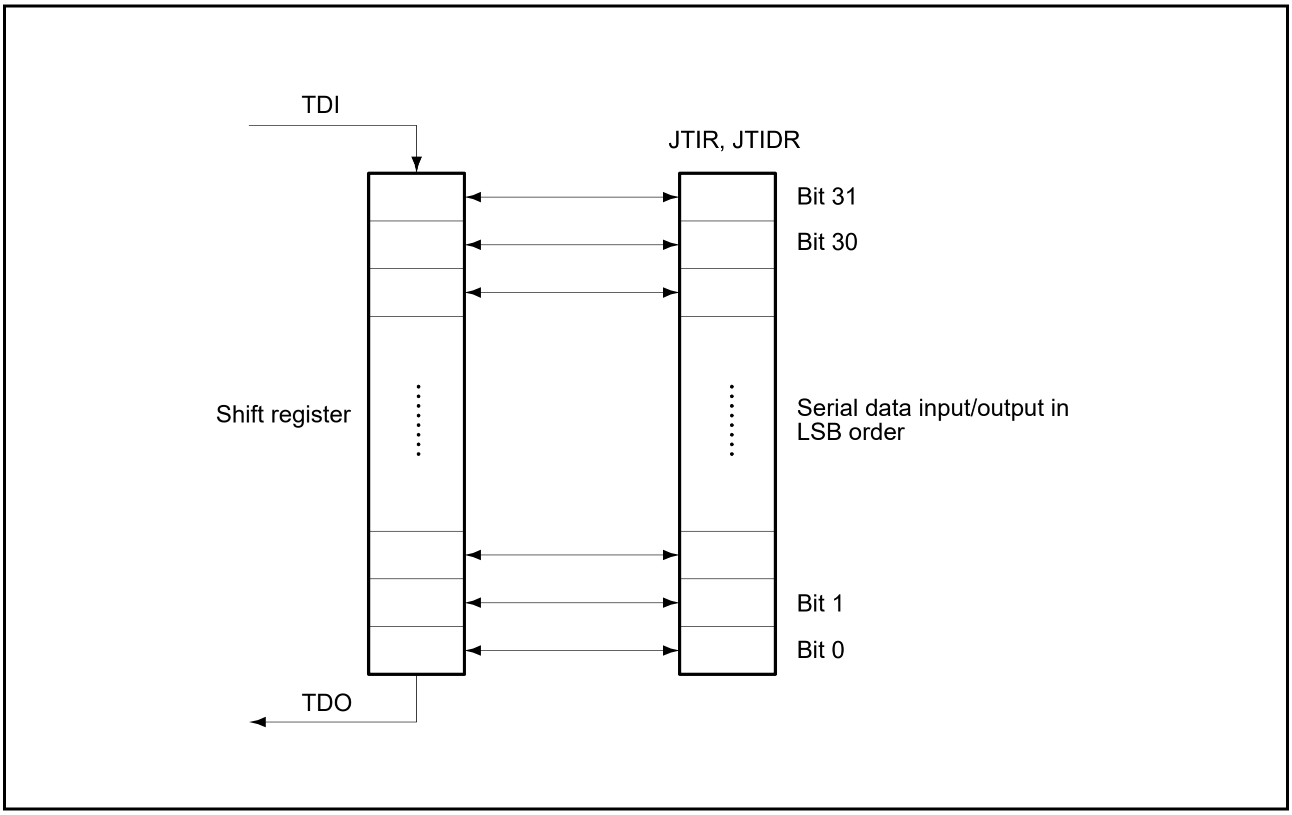


Figure 53.3 Serial Data Input/Output

(2) Pins of the boundary scan (TCK, TDI, TMS, and TRST#) have to be pulled up by pull-up resistors.
However, handle the TRST# pin in the way described in the manual for the given on-chip emulator if an on-chip emulator is in use.
If the TRST# pin is pulled down but a boundary scan is to proceed, ensure that the TRST# pin is also controllable.

(3) Power supply pins (VCC, VCL, VSS, AVCC0, AVCC1, AVSS0, AVSS1, VCC_USB, VSS_USB) cannot be boundary-scanned.

(4) Analog reference pins (VREFH0, VREFL0, VBATT) cannot be boundary-scanned.

(5) Clock pins (EXTAL, XTAL, XCIN, and XCOUT) cannot be boundary-scanned.

(6) Reset signal (RES#) cannot be boundary-scanned.

(7) USB dedicated pins (USB0_DP, USB0_DM) cannot be boundary-scanned.

(8) The on-chip emulator enable pin (EMLE) cannot be boundary-scanned.

(9) The boundary-scan pin (BSCANP) cannot be boundary-scanned.

(10) The boundary-scan pins (TCK, TMS, TRST#, TDI, and TDO) cannot be boundary-scanned.

(11) The boundary-scan facility is not available when the chip is in the states below.

- Reset state
- Software standby or deep software standby

(12) For a pin that incorporates open-drain functionality and for which the open-drain function is enabled, if the boundary-scan function sets the corresponding bit in the output scan register and output enable register to 1, executing an EXTEST, CLAMP, or SAMPLE/PRELOAD instruction makes the pin output the high level rather than placing it in the high-impedance state.

(13) Be sure to satisfy the standards for the boundary scan function when multiplex ports are used. Figure 53.4 (1) shows the pin configuration of the multiplex port pins in combination with the RIIC pins (P12, P13, P16, P17, P20,

and P21). When the boundary scan function is used with pins P12, P13, P16, P17, P20, and P21 to be used as RIIC pins (SCL0[FM+], SCL2, SDA0[FM+], and SDA2), the conflict with open-drain output or sneak current might be generated.

(14) Figure 53.4 (2) shows the pin configuration of the pins P00 to P02, P40 to P47, P90 to P93, PD0 to PD7, and PE0 to PE7. When the boundary scan function is used with pins P00 to P02, P40 to P47, P90 to P93, PD0 to PD7, and PE0 to PE7 to be used as AD input pins (AN000 to AN007, ANEX0, ANEX1, and AN100 to AN120), the conflict with the AD input or sneak current might be generated.

(15) Figure 53.4 (3) shows the pin configuration of pins P03 and P05. When the boundary scan function is used with pins P03 and P05 to be used as DA output pins (DA0 and DA1), the conflict with the DA output or sneak current might be generated.

(16) The HIGHZ mode option of the MD pin is not available.

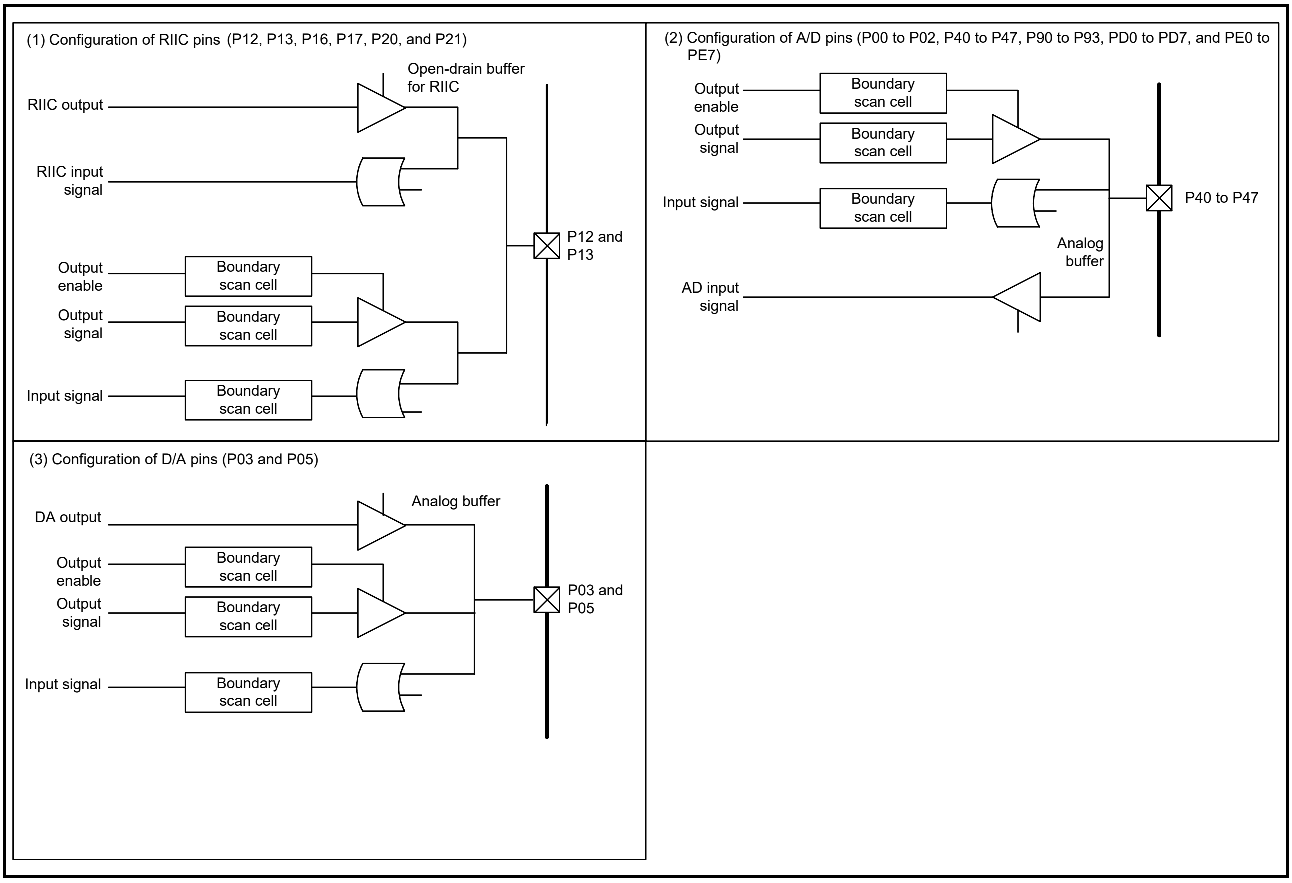


Figure 53.4 Pin Configuration